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EXPERIMENT-7

HTTP ANALYSIS USING WIRESHARK

Q) Start your web browser and clear the browser's cache memory. Open wireshark and start capturing. Then visit any webpage of your choice. Type http in the filter field of the Wireshark and click Apply so that only HTTP messages are displayed. After enough packets have been captured, select the 'Capture' from the pull down menu and select 'Stop Capturing'

Using the captured information, determine the following:

(a) the Source IP address and destination IP address of the first GET message

- Source IP address: 192.168.4.19
- Destination IP address: 192.168.34.90

(b) the medium format, the language, the encoding and the character set that the client can accept (Use the first GET message)

- Media format (Accept): text/html, application/xhtml+xml, application/xml, image/avif, image/webp, image/apng,**
- language (Accept-language): en-GB, en-US; q=0.9, en; q=0.8
- Encoding (Accept-encoding): gzip, deflate
- Character set: nil

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(c) the URL of the website & user agent (first GET message)

- URL: http://192.168.34.90/1
- User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/99.0.4844.82 Safari/537.36

(d) Source IP address and destination IP of first response

- Source IP address: 192.168.34.90
- Destination IP address: 192.168.4.19

(e) Status code of the first response message

- Status code: 200
- Status phrase: OK
- Complete status line: HTTP/1.1 200 OK

(f) When the HTML file that you are retrieving was last modified at the server

- Last-Modified: Sat, 14 Mar 2026 05:30:09 GMT

(g) Value of content-length field of first response message

- Content-Length: 2578 bytes

(h) Time taken between sending the GET request and receiving the response

- Time since request: 0.040339658 seconds

(i) the HTTP version of your browser

- HTTP version: HTTP/1.1

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Result : HTTP Analysis using wireshark was successfully implemented.

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EXPERIMENT - 8

DNS ANALYSIS USING WIRESHARK

Q) First, clear the DNS record from the cache memory of your computer. For this use ipconfig / flushdns on windows or systemd-resolve --flush-caches on Linux. Next, clear your browser's cache memory. Open the wireshark and start capturing. In your browser, visit your college website. Wireshark starts to capture packets. Type dns in the filter field and press Apply so that only DNS messages are displayed. Stop capturing and save the captured file

Using the captured information, answer the following questions:

(a) Locate the first DNS query message resolving your college website. What is the packet number in the trace for the DNS query message?

- Packet number: 146951

(b) Is the query sent over UDP or TCP

- Protocol: UDP

(c) Now locate the corresponding DNS response to the initial DNS query. What is the packet number in the trace for DNS response message? Is this response message received via TCP or UDP?

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- Packet number and protocol of corresponding DNS response
→ packet number: 146953
→ protocol: UDP

(d) What are the source and destination port numbers for the DNS query message?
- Source port: 45869
- Destination port: 53

(e) What are the source and destination port numbers of the DNS response message
- Source port: 53
- Destination port: 45869

(f) To what IP address is the DNS query message sent?
- Destination IP address: 192.168.34.254

(g) What is the query message ID number? What is the response message ID number? What is the purpose of this field?
- Query message ID: 0x7d5c
- Response message ID: 0x7d5c
- Purpose: The transaction ID identifies a DNS request and its corresponding response. The client uses this ID to match the response with original query.

(h) What is the length of the flag field in a DNS message
- Length: 16 bits (2 bytes)

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(i) Which bit is the flag field determines whether the message is a query or response?

- Bit: QR (Query / Response) bit, the first bit of the DNS flags field
- 0 indicates a query
- 1 indicates a response

(j) Which bits are used only in the response message? What is the function of these bits in the response?

→ AA (Authoritative Answer): indicates whether the responding server is authoritative for the queried domain

→ TC (Truncated): indicates that the response was truncated

→ RA (Recursion Available): indicates whether the server supports recursive queries

→ RCODE (Response Code): indicates the result of the query such as success or name error

(k) How many question records, answer records, authority records and additional records are present in the query message?

Question records: 1
Answer records: 0
Authority records: 0
Additional records: 0

(l) How many question records, answer records, authority records and additional records are present in the

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response message?

→ Question records : 1

. Answer records : 0

. Authority records : 0

. Additional records : 0

Result : The DNS traffic generated while accessing the college website was successfully captured and analysed using Wireshark.

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EXPERIMENT - 9

Familiarization of Router Commands

Aim

To familiarize with router modes, information commands, routing-table display, configuration saving, command history, clock hosts, interface statistics, and serial/Ethernet interface configuration

COMMANDS

(a) Knowing the current mode and switching to privileged mode

Router > mode

Router #

The 'enable' command is used to switch from the user EXEC mode to privileged EXEC mode.

(b) Switching to configuration mode

Router # configure terminal

The 'configure terminal' command is used to enter the global configuration mode

(c) Obtaining router information such as type, OS, memory, stats, interface details etc.

Router # show version

Router # show memory

Router # show interfaces

Router # show ip interface brief

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(d) Viewing the status of routing protocols:
Router# show ip protocols
This command displays information about routing protocols configured on the router

(e) Showing the router table
Router# show ip route
This command displays the routing table containing directly connected, static and dynamically learned routes

(f) Saving the running configuration
Router# copy running-config startup-config
This command uses the current running configuration to update and save the startup configuration so that it is retained after a reboot.

(g) Viewing the command history
Router# show history
This command displays previously entered commands in the current session

(h) Viewing the router clock
Router# show clock
This command displays the current date & time of the router

(i) Viewing the list of hosts
Router# show host

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This command displays the host names & their associated addresses stored on the router's host table

j) Display interface stastics
Router # show ip interface brief
This command displays a quick summary of the device's interfaces, including each interface's IP address and whether its status and line protocol are up or down

k) ~~Confi~~ Knowing the controller type (DTE or DCE)
Router #> show controllers serial 0/0/0
This command displays the controller details for the serial interface 0/0/0, including whether the cable end is DCE or DTE (that tells you which router end should provide clocking)

(l) Configuring serial and ethernet interfaces – enabling the interface, setting IP address mask and clock rate
ethernet
enable
configure terminal
interface gigabit Ethernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

serial
interface serial 0/0/0
ip address 10.0.0.1 255.255.255.252

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clock rate 64000
no shutdown
end

ethernet interface - assign an IP address and subnet mask
serial interface - assign IP and mask and set a clock rate only if router's serial end is DCE & enable it

Result : The basic Cisco packet tracer router commands were studied and practised.

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EXPERIMENT - 10

STATIC ROUTING

Q) A sample network along with the interface addresses are given (all interfaces use a /24 mask). Set up static routing for the network and once the routes are set up, display the routing table and verify the connectivity using ping.

Aim: To configure static routes for the network and display the routing table and finally verify connectivity using ping command using the Cisco packet tracer tool.

Algorithm

1. Open Cisco packet tracer and connect three Cisco 1941 routers (R1, R2, R3) in series using copper crossover cables between routers and straight through cables b/w PC and routers

2. Configure router interfaces using the following IP:

R1	G0/0	192.168.10.1
R1	G0/1	10.0.12.1
R2	G0/0	10.0.12.2
R2	G0/1	10.0.23.1
R3	G0/0	10.0.23.2
R3	G0/1	192.168.30.1

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3. Configure each interface using enable, configure terminal, interface, ip address and no shutdown

4. Configure static routes on each router

→ R1

ip route 192.168.30.0 255.255.255.0
10.0.12.2

→ R2

ip route 192.168.10.0 255.255.255.0
10.0.12.1
ip route 192.168.30.0 255.255.255.0
10.0.23.2

→ R3

ip route 192.168.10.0 255.255.255.0
10.0.23.1
Enter routes in global configuration mode. Then use end and copy-running-config startup-config on each router

5. Configure PC IP settings

→ PC0: IP 192.168.10.10, mask 255.255.255.0, gateway 192.168.10.1

→ PC1: IP 192.168.30.10, mask 255.255.255.0, gateway 192.168.30.1

6. Verify interfaces and routing tables using show ip interface brief and show ip route and test

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PC command prompt using ping 192.168.30.10 on PC0 and ping 192.168.10.10 on PC1

Result: Static routing was configured and verified using routing tables and ping tests.

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EXPERIMENT - 11

RIPV2 ROUTING

Q) Implement RIPv2 routing for the network shown once the routes are setup, display the routing table and verify its connectivity using ping.

Aim: To implement RIPv2 routing for the network and verify connectivity using Cisco packet tracer

Algorithm

1. Open Cisco Packet Tracer and use three cisco routers R1, R2 and R3 and connect them using copper-crossover cables and PCs using straight-through cables

2. Configure the router interfaces with following IP addresses

Router	Interface	IP address	subnet mask
R1	G0/0	192.168.10.1	255.255.255.0
R1	G0/1	10.0.12.1	255.255.255.252
R2	G0/0	10.0.12.2	255.255.255.252
R2	G0/1	10.0.23.1	255.255.255.252
R3	G0/0	10.0.23.2	255.255.255.252
R3	G0/1	192.168.30.1	255.255.255.0

3. Enable the interfaces using no shutdown and verify with show ip interface brief.

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4. Configure PC0 as 192.168.10.10/24, gateway: 192.168.10.1
PC1 as 192.168.30.10/24, gateway 192.168.30.1

5. Configure RIPv2 on each router

R1: router rip
version 2
no auto-summary
network 192.168.10.0
network 10.0.0.0
end

R2: router rip
version 2
no auto-summary
network 10.0.0.0
end

R3. router rip
version 2
no auto-summary
network 10.0.0.0
network 192.168.30.0
end

6. Verify RIP and routing tables on all routers using show ip protocols and show ip route

7. Test connectivity from PC command prompt using ping 192.168.30.10 on PC0 and ping 192.168.10.10 on PC1
Confirm successful replies

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Result : RIPv2 routing was configured, routing tables were displayed and connectivity was verified using ping.

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EXPERIMENT - 12

OSPF Routing

Q) Implement OSPF routing for the network. Once the routes are set up, display the routing table and verify the connectivity using ping.

Aim: To implement OSPF routing for the network and verify connectivity using Cisco Packet tracer tool

Algorithm

1. Open Cisco Packet tracer and create topology using 3 Cisco routers R1, R2 and R3 and two PCs: PC0, PC1 Connect routers using copper-cross over cables & PCs using straight through cables.

2. Configure router interfaces with following IP:

Router	Interface	IP address	subnet mask
R1	G0/0	192.168.10.1	255.255.255.0
R1	G0/1	10.0.12.1	255.255.255.252
R2	G0/0	10.0.12.2	255.255.255.252
R2	G0/1	10.0.23.1	255.255.255.252
R3	G0/0	10.0.23.2	255.255.255.252
R3	G0/1	192.168.30.1	255.255.255.0

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3. Enable the interfaces using no shutdown and verify using show ip interface brief and configure PC0 with IP 192.168.10.10/24 gateway 192.168.10.1 and PC1 with IP 192.168.30.10/24 gateway 192.168.30.1

4. Configure OSPF on each router

R1
no router rip
router ospf 1
router-id 1.1.1.1
network 192.168.10.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.3 area 0
end

R2
no router rip
router ospf 1
router-id 2.2.2.2
network 10.0.12.0 0.0.0.3 area 0
network 10.0.23.0 0.0.0.3 area 0
end

R3
no router rip
router ospf 1
router-id 3.3.3.3
network 10.0.23.0 0.0.0.3 area 0
network 192.168.30.0 0.0.0.255 area 0
end

5. Verify OSPF neighbours and routing tables on all routers
show ip ospf neighbour and show ip route

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6. Test connectivity from PC command prompts using PC0: ping 192.168.30.10 and on PC1 ping 192.168.10.10

Result: OSPF routing was configured, neighbour relationships and routing tables were verified and connectivity was tested using ping.

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EXPERIMENT – 13

Campus Network Access Control using ACL

Q) You are the network administrator of your college A small portion of your campus network is shown in figure. You want to allow only Host_B to communicate with the network 172.16.10.0. Verify your settings by the following checks

(a) Pinging Host_A and Host_B

(b) Pinging Host_A from lab_B and lab_C

Aim : To configure access control so that only Host_B can communicate with the 172.16.10.0 network and verify the required connectivity tests using Cisco packet tracer

Algorithm

1. Open the Cisco packet tracer and recreate the network using suitable routers, PCs and switches.

2. Configure IP addresses of hosts & router interfaces according to the topology. The Host

Host_A network : 172.16.10.0/24

Host_B network : 172.16.30.0/24

Host A IP : 172.16.10.10

Host B IP . 172.16.30.10

Subnet mask : 255.255.255.0

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3. Configure the router interface connected to Host_A network
interface gigabitEthernet 0/0
ip address 172.16.10.1 255.255.255.0
no shutdown

4. Create an extended ACL named HOSTB_ONLY to allow Host_B and deny other sources from accessing the Host_A network
ip access-list extended HOSTB_ONLY
permit ip host 172.16.30.10 172.16.10.0 0.0.0.255
deny ip any 172.16.10.0 0.0.0.255
permit ip any any
exit

5. Apply ACL outbound on interface connected to Host_A LAN and save the configuration

6. Verify the ACL configuration using show access-lists, show ip interface brief and show running-config

7. Test connectivity using command-prompts
From Host_B: ping 172.16.10.10 - success
From Lab_B : ping 172.16.10.10 - fail
From Lab_C : ping 172.16.10.10 - fail

Result: An extended ACL was configured to allow Host_B to access the 172.16.10.0/24 network while denying other sources and connectivity was verified using ping.

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EXPERIMENT - 14

Extended ACL - Blocking a specific Website From CCF

Q) You are the network administrator of your college. The college is assigned a network address 140.80.0.0 There are 20 subnets in your college network. The Central Computing Facility (CCF) resides in the 4th subnet. The department of CSE is organizing an inter-department hackathon for which the registration closed yesterday. The registration was through the hackathon website hosted on a server which is assigned the 7th address in the 16th subnet. As the network administrator, your job now is to block students from accessing the hackathon website from CCF [The server provides other services than the website hosting as well. Make sure to block only the website access. other services should not be denied)

Aim : To use an extended ACL to block students in CCF from accessing the specified hackathon website while allowing other services from the server using cisco packet tracer

Algorithm

1. Open Cisco Packet Tracer and create the college network topology with routers, switches, CCF network and the hackathon web server

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2. Subnet the network 140.80.0.0 to obtain 20 subnets using /21 subnet mask (255.255.248.0), identify the required networks

- CCF – 4th subnet : 140.80.24.0/21
- Server – 16th subnet : 140.80.120.0/21
- Server IP (7th usable address) : 140.80.120.7

3 Configure router interfaces and assign IP to hosts according to the topology and ensure that interfaces are enabled using no shutdown

4. Configure an extended ACL to block only HTTP and HTTPS traffic from CCF to server

```
ip access-list extended
BLOCK_WEBSITE
deny tcp 140.80.24.0 0.0.7.255 host
140.80.120.7 eq 80
deny tcp 140.80.24.0 0.0.7.255 host
140.80.120.7 eq 443
permit ip any
exit
```

5. Apply the ACL inbound on the router connected to the CCF network

6. Verify ACL configuration using show access-lists and show running-config.

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7. Test website access from a CCF host. HTTP and HTTPS will be blocked while other services on the server and traffic to other networks will remain permitted.

Result: An extended ACL was configured to block only website access from CCF to the hackathon server while allowing other services.